

Logistic Regression From scratch:

Input = X^i (N features) (x^0 for bias b) (S samples)

parameters = Θ (n parameters) (bias b included)

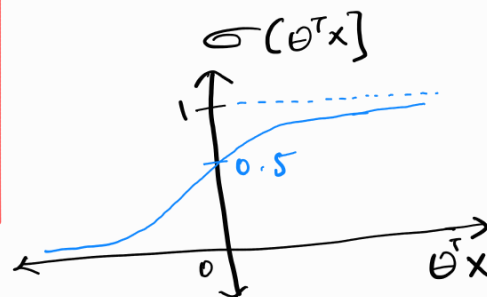
$$\hat{Y} = \Theta^T X$$

$$\left[\begin{array}{l} \text{dimensions} \\ \Theta_{N \times 1} \\ X_{N \times S} \end{array} \right] \Rightarrow \left[\begin{array}{l} \therefore \Theta^T X \text{ OR } X^T \Theta \\ \downarrow \text{transpose} \\ Y_{S \times 1} \end{array} \right]$$

Now to introduce non linearity we use sigmoid function.

$$\hookrightarrow \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\therefore \sigma(y) = \sigma(\Theta^T x) = \frac{1}{1 + e^{-\Theta^T x}}$$



↳ sigmoid squashes the output between $(0, 1)$

↳ we can interpret this squashed values as probabilities.

↳ we can use threshold for binary classification.

why 0.5?

$$\hookrightarrow \frac{1}{1 + e^{-\theta^T x}} = y$$

$$y = \frac{1}{1 + e^{-0}} \quad (\theta^T x = 0)$$

$$y = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

we need to get optimal parameters.

i.e. Error $(\theta, \bar{\theta}) \rightarrow 0$
Loss
actual parameters $\swarrow$
predicted parameters $\searrow$

$$L(\theta) = \prod_{i=1}^S [\sigma(\theta, x^i)]^{y^i} [1 - \sigma(\theta, x^i)]^{1-y^i}$$

verify: since we can never reach 1/0

① $\sigma(\theta, x^i) \rightarrow 1$ $\leftarrow$ predicted y^i
 $y^i = 1$ $\leftarrow$ actual y^i

② $\sigma(\theta, x^i) \rightarrow 0$ $\leftarrow$ predicted y^i
 $y^i = 0$ $\leftarrow$ actual y^i

$$L(\theta) = \prod_{i=1}^S (1)^1 \cdot (1-1)^{1-1}$$

$$L(\theta) = \prod_{i=1}^S (0)^0 \cdot (1-0)^{1-0}$$

$L(\theta) = 1$ $\searrow$ $(0^0 = 1)$
assume

$L(\theta) = 1$

why this formula?

- $(\sigma(\theta))^{y_i}$ selects prediction when target is 1
- $(1 - \sigma(\theta))^{1-y_i}$: selects when target is 0.
- Π : product because we consider all instances independent.

likelihood function $L(\theta)$

$\searrow$ tells how well parameters θ explain the observed data x .

$\therefore$ we need to maximize this likelihood.

lets consider 1000s of instances how will $L(\theta)$ react

$$L(\theta) = \prod_{i=1}^n (\sigma(x_i))^{y_i} (1 - \sigma(x_i))^{1-y_i}$$

$\hookrightarrow$ product of 1000s of numbers between (0-1) will give us very very small numbers.

$\therefore$ take log to convert product to +

$$\therefore LL(\theta) = y_i \log(\sigma(x_i)) + (1 - y_i) \log(1 - \sigma(x_i))$$

$\downarrow$
we need to maximize this
OR

we can minimize this } in this case we can use
"Gradient descent"

$\therefore -$

$$\therefore LL(\theta) = -y_i \log(\sigma(x_i)) - (1 - y_i) \log(1 - \sigma(x_i))$$

$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n (LL(\theta))$$

[Binary cross entropy]
[cost function]

Gradient descent:

$$\theta = \theta - \lambda \frac{\partial J(\theta)}{\partial \theta}$$

$$\frac{\partial J(\theta)}{\partial \theta} = \sum_{i=1}^n \left(-y_i \log(\sigma(x_i)) - (1-y_i) \log(1-\sigma(x_i)) \right)$$

$$= \frac{-y_i}{\sigma(x_i)} \cdot \frac{\partial \sigma(x_i)}{\partial \theta} - \frac{(1-y_i)}{(1-\sigma(x_i))} \cdot \left[-\frac{\partial \sigma(x_i)}{\partial \theta} \right]$$

$$= \frac{-y}{\left(\frac{1}{1+s}\right)} \cdot \frac{-sx}{(1+s)^2} + \frac{1-y}{\left(1-\frac{1}{1+s}\right)} \cdot \left[\frac{-sx}{(1+s)^2} \right]$$

$$= \frac{+ysx}{1+s} - \frac{1-y}{s} \cdot \frac{sx}{1+s}$$

$$= \frac{x}{1+s} \left[ys - (1-y) \right]$$

$$= \frac{x}{1+s} (ys - 1 + y)$$

$$= \frac{x}{1+s} (y(1+s) - 1)$$

$$= xy - \frac{x}{1+s}$$

$$= \left(y - \frac{1}{1+s} \right) x$$

$$= (y_{\text{actual}} - y_{\text{pred}}) x$$

$$\left[\sigma(x_i) = \frac{1}{1+e^{-\theta x_i}} \right]$$

$$s = e^{-\theta x_i}$$

$$\frac{\partial \sigma(x_i)}{\partial \theta} = \frac{-e^{-\theta x_i} \cdot x_i}{(1+s)^2}$$

$$= \frac{-sx}{(1+s)^2}$$